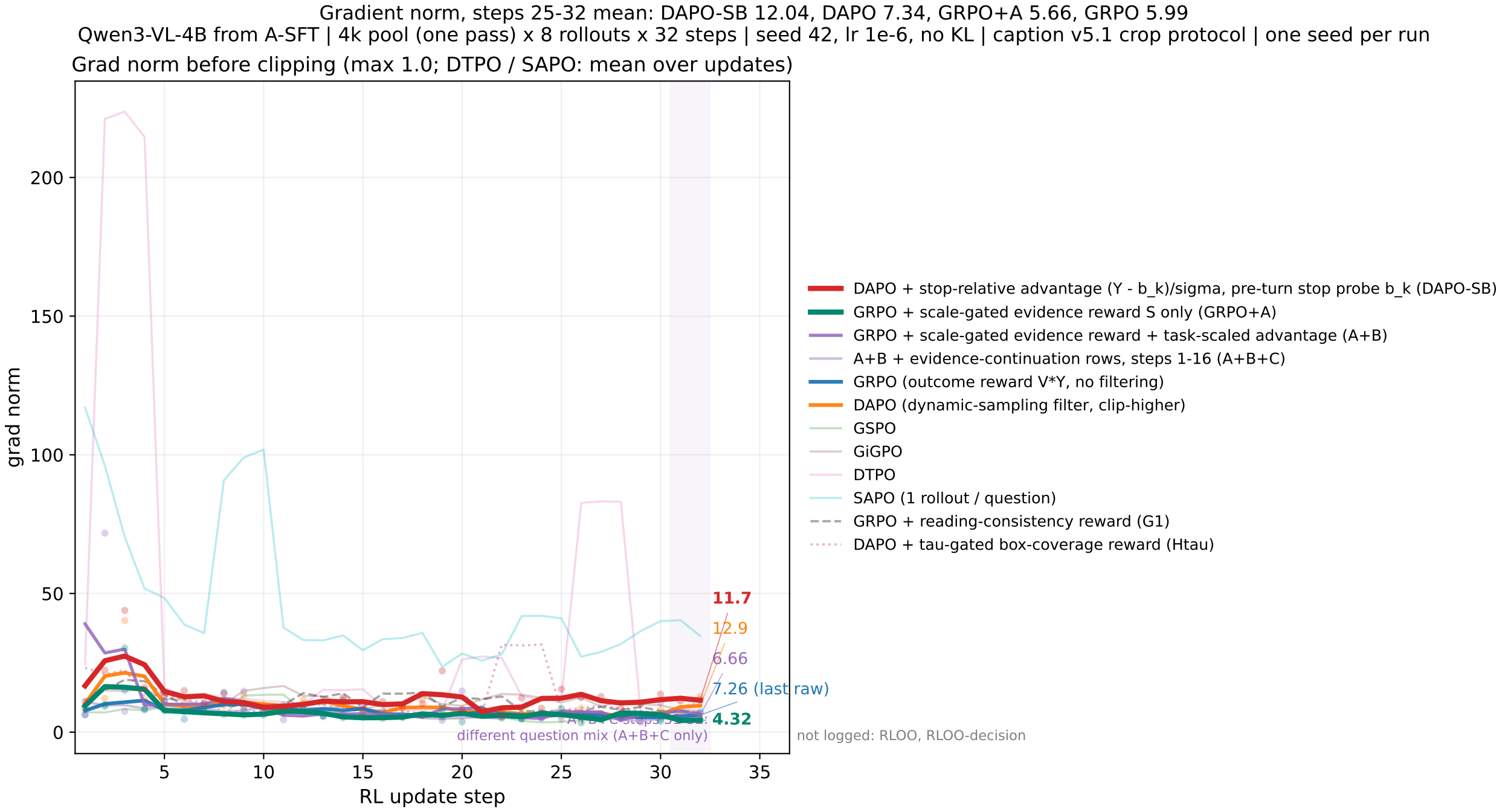
Gradient norm, steps 25-32 mean: DAPO-SB 12.04, DAPO 7.34, GRPO+A 5.66, GRPO 5.99
Qwen3-VL-4B from A-SFT | 4k pool (one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, no KL | caption v5.1 crop protocol | one seed per run
Grad norm before clipping (max 1.0; DTPO / SAPO: mean over updates)
grad norm
RL update step
200
150
100
50
0
5
10
15
20
25
30
35
11.7
12.9
6.66
7.26 (last raw)
4.32
different question mix (A+B+C only)
DAPO + stop-relative advantage (Y - b_k)/sigma, pre-turn stop probe b_k (DAPO-SB)
GRPO + scale-gated evidence reward S only (GRPO+A)
GRPO + scale-gated evidence reward + task-scaled advantage (A+B)
A+B + evidence-continuation rows, steps 1-16 (A+B+C)
GRPO (outcome reward V*Y, no filtering)
DAPO (dynamic-sampling filter, clip-higher)
GSPO
GiGPO
DTPO
SAPO (1 rollout / question)
GRPO + reading-consistency reward (G1)
DAPO + tau-gated box-coverage reward (Htau)
not logged: RLOO, RLOO-decision